



**Carisurg**

Assignment 12

Final Proposal

**Due Date:** July 3rd, 2026.

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## Problem Statement

Emergency departments in resource-constrained Caribbean settings like Mercer General Hospital operate on paper-based triage systems that generate no structured, queryable data. Without a digital foundation, locally validated AI triage tools cannot be built, and globally trained models — developed predominantly on North American and European patient populations — cannot be reliably applied. This project proposes to establish a structured digital triage dataset from the Mercer General ED as the necessary first step toward a clinically credible, regionally relevant AI-assisted triage classifier.

## Project Context

This project is structured as two phases, referenced repeatedly throughout the proposal, diagrams, and weekly write-ups. They are not interchangeable — each has a different scope, timeline, and deliverable.

**Phase 1 — the 12-week pilot (this programme).** The goal is to design and validate a structured **digital triage capture tool** that mirrors the existing Mercer General paper triage form. The output is not an AI model — it is a clean, de-identified, queryable dataset built from real ED encounters. Mercer currently runs entirely on paper, which means no structured triage data exists anywhere to train or validate a model against. Phase 1 exists to fix that gap first.

**Phase 2 — beyond this programme (future work).** Once a sufficient digital dataset exists from Phase 1, Phase 2 would use that data to train and locally validate a **rule-based ESI classifier**, iterating toward a machine learning model as data volume grows over time. This is the long-term

vision the project is building toward — but it is explicitly *not* a deliverable of the current 12-week pilot.

**Why the split matters:** Most AI-assisted triage literature assumes infrastructure — electronic health records, stable data pipelines — that Mercer does not have. Proposing an AI classifier without first proposing the data foundation it depends on would be building on top of nothing. Phase 1 is the foundation. Phase 2 is the building.

## **Previous Works & How we Move Forward**

Research into AI and machine learning in ED triage has grown significantly, with studies showing promise in earlier diagnosis and intervention, though concerns remain around overconfident outputs presenting risks to patients (Tyler et al., 2024). AI-driven triage systems have demonstrated potential by automating patient prioritisation through real-time analysis of vital signs, medical history, and presenting symptoms, with narrative reviews covering literature from 2015 to 2024 identifying both benefits and persistent limitations (Da’Costa et al., 2025).

A systematic review of prospective studies found that while AI exhibits potential in emergency patient triage, it has not yet established evidence for substituting human involvement — a critical distinction for a setting like Mercer, where nurse-led triage is the primary model and clinician trust is a known adoption barrier (Yi et al., 2025).

Infrastructure constraints, including unreliable power supply, limited internet connectivity, and the absence of EHR integration tools, alongside a lack of high-quality annotated datasets for specific populations, represent key barriers to AI triage implementation in resource-limited

settings. These barriers are directly applicable to the Caribbean context. A recent scoping review examining AI triage across low- and middle-income countries found that 72% of studies were conducted in high-income countries, with LMICs representing only 28% of the evidence base — reflecting both the thinness of the evidence base in resource-constrained settings and the near-complete absence of Caribbean data from this literature (Aifuobhokhan et al., 2025).

Araouchi & Adda (2024) demonstrated a multimodal AI triage system integrating structured vitals with unstructured clinical text, showing accuracy improvements over single-modality approaches — but validated exclusively on datasets from high-income settings with complete electronic health records, a precondition Mercer does not currently meet.

Where machine learning models have been developed and internally validated in non-Western settings, results are promising but context-dependent. Sitthiprawiat et al. (2025) developed an XGBoost model across 163,452 ED visits at a Thai hospital, achieving superior predictive performance over the Canadian Triage and Acuity Scale (AUROC 0.917 vs. 0.882) — but noted that generalisability to different patient populations and healthcare infrastructures remains unproven, underscoring the need for locally trained models in every distinct clinical context.

Critically, there is precedent for the Phase 1 approach proposed here. Mitchell et al. (2024) documented triage implementation at Vila Central Hospital in Vanuatu — a Pacific Small Island Developing State with infrastructure constraints directly analogous to the Caribbean — demonstrating the feasibility of a low-cost electronic data registry system built to capture triage data in a previously paper-based ED. Their key finding was that a bespoke needs assessment of local triage requirements was essential before any digital tool could be designed, a lesson directly applicable to the Mercer context.

De Freitas et al. (2020) conducted a qualitative exploration of patient flow in a Caribbean emergency department, directly documenting the operational realities — overcrowding, delays, and resource constraints — that any digital intervention at Mercer must be designed around. This paper anchors the workflow assumptions made in the systems-thinking analysis below in real Caribbean clinical observation rather than inference from non-Caribbean settings.

### **Deployment & Constraint-Focused Literature**

This week's literature additions shift focus from triage accuracy toward deployment reality — the gap between a model performing well in a paper and a model surviving contact with an actual clinical workflow.

Poon et al. (2025) surveyed 43 US health systems on AI adoption during the early generative AI era and identified six recurring barriers: financial concerns, regulatory uncertainty, lack of leadership support, low clinician adoption, insufficient expertise or technology, and immature AI tools. Notably, low clinician adoption was identified as a barrier independent of model accuracy — reinforcing that a technically sound tool can still fail if it does not fit how clinicians actually work.

Salwei et al. (2024) evaluated a human-factors-based clinical decision support (CDS) tool specifically within an emergency department and found that although usability testing scored highly in both experimental and real clinical settings, actual use of the tool was low. The dominant barrier identified was a lack of workflow integration — in one documented case, an EHR system upgrade silently relocated the tool within the interface, and because the change was

not communicated to physicians, it significantly disrupted their workflow and further suppressed adoption. This is a direct, real-world illustration of Constraint 1 below: a tool's clinical value collapses the moment it adds friction to an already time-pressured workflow.

Chughtai and Blanchet (2017) contributed a meta-narrative review establishing systems thinking as a methodology for public health interventions, arguing that health system interventions must be understood as embedded within a web of interacting actors, incentives, and constraints rather than as isolated technical fixes. This paper provides the methodological grounding for the workflow-mapping and constraint-identification approach taken in this proposal.

### **New This Week: Ethics, Safety & Equity Literature**

This week's additions shift focus toward the ethical and equity dimensions of AI deployment — the question of who is protected and who is exposed when a clinical AI system fails.

Obermeyer et al. (2019) conducted a landmark analysis of a widely deployed US healthcare risk-prediction algorithm, finding that it systematically under-identified Black patients for additional care management. The bias did not arise from an overtly discriminatory input, but from the algorithm's use of historical healthcare cost as a proxy for healthcare need — a proxy that itself reflected unequal access to care along racial lines. Correcting the proxy more than doubled the number of Black patients flagged as needing additional care. The lesson for this project is structural: even a model trained on entirely accurate, well-intentioned data can encode harmful bias through the choice of proxy variable, not just through the choice of training population. This directly informs Risk 2 and Risk 9 in the register below — a Caribbean training

dataset is necessary but not sufficient; the variables and proxies used within it require the same scrutiny.

The World Health Organization (2021) published comprehensive ethics and governance guidance for AI in health, developed over eighteen months by an expert group spanning ethics, digital technology, law, and human rights. The guidance establishes six consensus principles, foremost among them that human agency must remain paramount and that AI in health must operate within a human-in-the-loop or human-on-the-loop architecture — explicit AI assistance, never autonomous clinical authority. This guidance directly underwrites the design stance taken throughout this proposal: the Phase 2 classifier is framed as decision support, never decision replacement, and Risk 6 in the register below names automation bias as a named, monitored risk rather than an assumed non-issue.

Wong et al. (2021) and the accompanying editorial commentary by Habib, Lin, and Grant (2021) document the external validation failure of the Epic Sepsis Model, a deployed clinical early-warning tool whose real-world performance fell far short of its developer's internal claims once independently tested outside its original training context. This case forms the basis of the harm case study accompanying this proposal (committed separately as [docs/ai-harm-case-study.md](#)) and directly informs Risks 1, 4, and 9 in the register below.

## Workflow Notes

The Mercer General Emergency Department follows a door-to-disposition pathway with five measured timestamps: **arrival**, **triage**, **seen-by-physician**, **disposition decision**, and **exit**.

| Stage                    | Duration | What Happens                                                                                                        | Data Captured                   |
|--------------------------|----------|---------------------------------------------------------------------------------------------------------------------|---------------------------------|
| <b>Arrival</b>           | —        | The patient arrives via ambulance, walk-in, or private vehicle                                                      | None yet                        |
| <b>Registration</b>      | 2–3 min  | The patient’s data, including their demographics, consent, and arrival modality, is collected.                      | Front desk — paper intake form  |
| <b>Triage Assessment</b> | 3–5 min  | The patient’s severity is assessed based on chief complaint, vitals, allergies, pain score, and ESI level assigned. | Triage nurse — paper front page |
| <b>Zone Placement</b>    | —        | The patient is placed in the ESI zone. Zones 1–3 equate to bed placement (resus/acute); while                       | None — disposition only         |

|                             |                                        |                                                                                                                |                                   |
|-----------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------|
|                             |                                        | zones 4–5 equate to fast-track waiting.                                                                        |                                   |
| <b>Physician Assessment</b> | ESI 2: ≤10 min<br><br>ESI 3: 30–60 min | The physician accesses patient, collates their medical history, does examinations, and places orders for same. | Physician — deeper clinical pages |
| <b>Investigations</b>       | 30 min – 3 hr                          | The labs/imaging are processed, collected and accessed — <b>shared hospital resource.</b>                      | Lab/radiology system              |
| <b>Disposition Decision</b> | At the end of the workup               | The patient is either admitted/discharged/transferred based on the recommendation by the physician.            | Physician                         |
| <b>Exit</b>                 | Variable                               | The patient leaves the hospital; however, this is delayed if admitted.                                         | —                                 |

**Key observations on information loss:**

1. The triage front page (ESI, vitals, chief complaint) is **consistently completed**.
2. The deeper clinical pages (history, medication chart, disposition rationale) are **not consistently completed** — this is the first place structured data goes missing.
3. **Re-triage** depends entirely on a nurse visually scanning the waiting room at intervals. There is no structured trigger for re-assessment — a deteriorating ESI-4 patient is only caught if someone happens to notice.
4. Investigations are the dominant flow bottleneck. Lab and radiology are shared with the rest of the hospital, and ED workups compete directly with elective inpatient demand.

## Workflow Diagram

In the diagram below, decision diamonds are placed at the triage-level ESI assignment step, per task instructions. Along with five annotated AI plug-in points, including one explicit **non-plug-in point** — a place where adding AI would actively make things worse.

## **AI Plug In-Key**

### **1. Digital Triage Capture Tool**

Reduces the paper form with structured digital intake at Registration/Triage. This project, Phase 1.

### **2. ESI Decision Support**

Suggests an ESI level alongside the nurse's own judgement at the zone-placement decision point. This describes phase 2, future work.

### **3. Re-triage Alert**

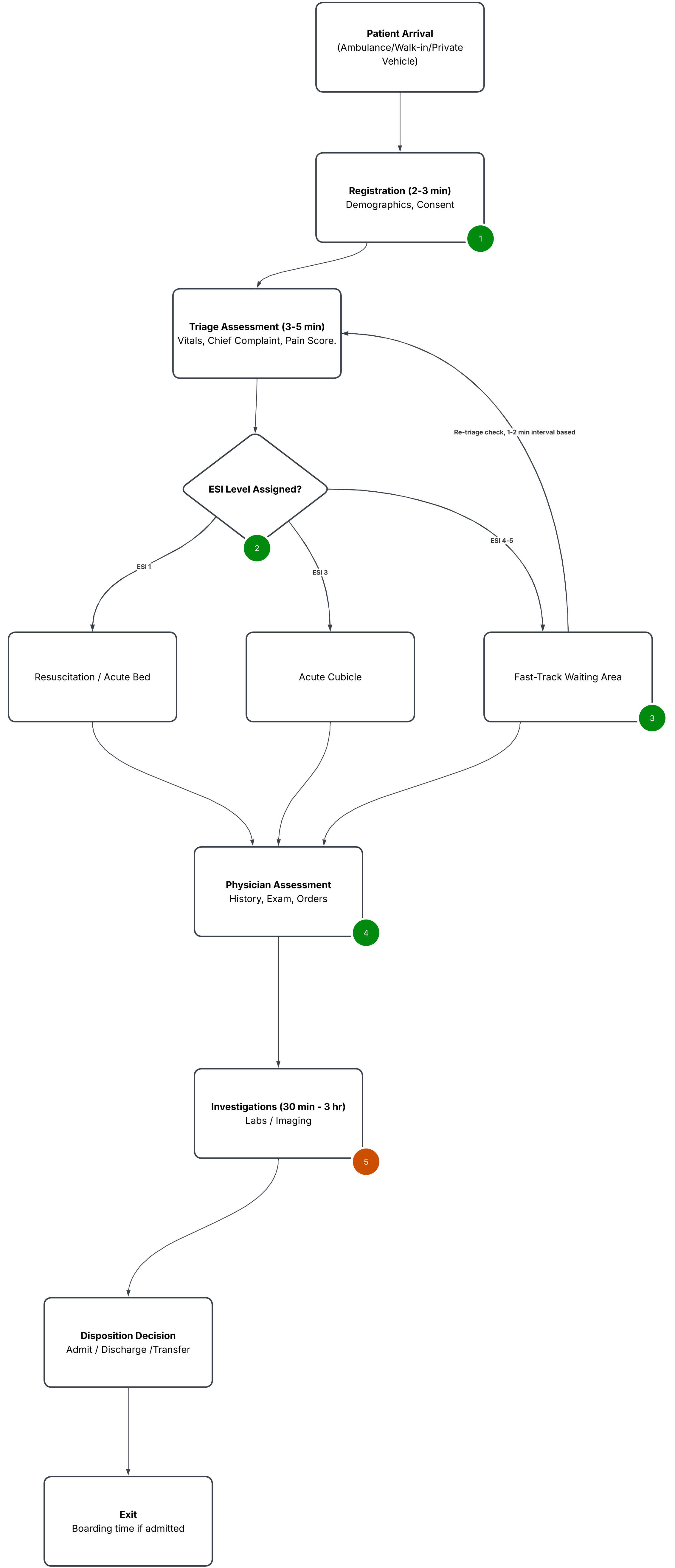
Flags overdue waiting room re-checks based on elapsed wait time and initial vitals

### **4. Structured Handover Prompt**

Auto-populates on SBAR-style summary from digital triage data at the physician handoff.

### **5. NOT a plug-in point — Investigations Bottleneck**

AI must NOT increase test-ordering volume without addressing lab/radiology turnaround. Doing so would worsen the queue, not relieve it.



## **Workflow Constraints**

### **Constraint 1 — The 3–5 Minute Triage Window (Friction Budget)**

Triage nurses have a narrow assessment window and a queue of waiting patients behind every encounter. Any digital tool that asks for more fields than the existing paper form, or takes more time than handwriting, will be abandoned within days — regardless of how good the underlying model is. This directly shapes the interface design for the Phase 1 capture tool: it must replicate the paper form's speed, not add to it.

### **Constraint 2 — Paper-to-Digital Lag at the Point of Capture**

All triage data currently exists only on paper, and even there, only the front page (ESI, vitals) is reliably completed. Before any classifier — rule-based or ML — can be trained or validated, structured digital data must exist at the moment of capture, not as a later transcription step prone to loss and delay. This is the foundational justification for the entire Phase 1 proposal.

### **Constraint 3 — Shared Investigations Bottleneck**

Labs and radiology are shared hospital-wide resources with 30-minute to 3-hour turnaround, and ED workups compete directly with elective inpatient demand. This is the dominant flow bottleneck at Mercer. Any AI design that increases test-ordering volume without addressing turnaround capacity will worsen overcrowding rather than relieve it — this constraint actively rules out certain plug-in points (see Investigations stage in diagram above) rather than just shaping their design.

## Clinical Stakeholders

The **five identified clinical stakeholders** at Mercer General, each with a distinct top concern that any triage tool should respect are:

| Stakeholder                          | Role Category      | Top Concern                                                                                                                        |
|--------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Triage Nurse                         | Nurse              | They make the right ESI call quickly, and attend to patients in the waiting room who have just gotten worse.                       |
| ED Physician /<br>Consultant on Call | Consultant         | They handle the workload and documentation burden. They need decision support that doesn't slow them down or add to charting time. |
| EMS Crew                             | Porter / Transport | They need a clean, SBAR-style handover.<br>Additional waiting hours to clear their stretcher for the next call impede them.        |
| Registration Clerk                   | Admin              | They need to get demographics right at a moment when the patient may be frightened, unwell, or unable to communicate clearly.      |

|                  |                  |                                                                                                                                |
|------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Patient & Family | Patient Advocate | They need to be informed of what is happening,<br>Especially when activity in the department appears<br>slow from the outside. |
|------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------|

### **Do We Consider the Porter Role and Do We Consider It?**

From the reading, Mercer General does not have a dedicated porter position documented in the available operational brief. Ergo, EMS crews are used here as the closest functional equivalent. They're responsible for a physical handoff into the department, with a comparable concern around minimising delay at that handoff point.

### Risk Register Table

This register identifies 10 named risks spanning the AI-technical, operational, ethical, and equity dimensions of a phased AI-assisted triage rollout, modelled on the Phase 1 (digital capture) / Phase 2 (rule-based and ML classifier) approach proposed for Mercer General ED. Each risk includes likelihood, impact, a proposed mitigation, and a measurable signal that the mitigation is actually working — not just present on paper.

| # | Risk                                                                                                                                                                | Category     | L | I | Mitigation                                                                                                       | Signal of Success                                                                                                      |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|---|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| 1 | Distribution shift / single-site overfitting. A model trained on Mercer's patient population performs poorly when deployed at a different hospital with a different | AI-technical | H | H | Mandate site-specific external validation before any new deployment; no “train once, deploy everywhere” rollout. | AUC/sensitivity tracked per site quarterly; deployment paused if performance drops below a pre-agreed local threshold. |

|   |                                                                                                                                                                                                    |              |   |   |                                                                                                                                            |                                                                                                                               |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---|---|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|   | demographic and disease mix.                                                                                                                                                                       |              |   |   |                                                                                                                                            |                                                                                                                               |
| 2 | Algorithmic bias from non-representative training data. No Caribbean ED triage dataset currently exists; bootstrapping from non-Caribbean data risks encoding assumptions that don't hold locally. | AI-technical | H | H | Phase 1 deliverable is exclusively Caribbean-sourced data; no transfer learning from non-Caribbean models without full local revalidation. | Published demographic audit comparing training set composition against the ED's actual catchment population.                  |
| 3 | Silent failure — no uncertainty signal on ESI suggestions. A classifier outputs a triage suggestion with no indication of how                                                                      | AI-technical | M | H | Every suggestion must surface the triggering rule or confidence score alongside the ESI level — never a bare number.                       | 100% of system-generated suggestions are paired with an explainable trigger or confidence indicator (audited at code review). |

|   |                                                                                                                                                        |             |   |   |                                                                                                                                                               |                                                                                                                                                      |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | confident or borderline it is.                                                                                                                         |             |   |   |                                                                                                                                                               |                                                                                                                                                      |
| 4 | Alert fatigue from over-triage. High false-positive volume causes nurses to start ignoring or dismissing alerts reflexively, including true positives. | Operational | H | H | Conservative initial alert thresholds; minimum locally-validated positive predictive value required before go-live; nurse-tunable thresholds post-deployment. | Alerts-per-true-positive ratio tracked monthly; target materially better than the 109:1 ratio documented in the Epic Sepsis Model case (see Part 2). |
| 5 | Workflow friction is causing data-entry shortcuts. Nurses copy-paste from the previous patient or skip fields if the digital tool                      | Operational | H | M | Phase 1 fields are capped at what the existing paper form already captures; usability testing confirms $\leq 5$ -minute completion time.                      | Field completion audit; target $>95\%$ of entries are genuine (non-default, non-duplicated) values.                                                  |

|   |                                                                                                                                                           |         |   |   |                                                                                                                                       |                                                                                                                                                               |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---|---|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | takes longer than the paper form.                                                                                                                         |         |   |   |                                                                                                                                       |                                                                                                                                                               |
| 6 | Automation bias. Nurses defer to the system's suggested ESI level even when it contradicts their own clinical judgment, particularly under time pressure. | Ethical | M | H | System is framed explicitly as decision support, never decision maker; any override requires the nurse to document a one-line reason. | Override rate tracked over time.<br><br>Caveat: a near-zero override rate is NOT success — it may indicate automation bias. Needs a paired qualitative audit. |
| 7 | Informed consent gap. Patients are triaged, and their data feeds a digital/AI system without being meaningfully                                           | Ethical | M | M | Visible signage and a short verbal consent disclosure at registration; an opt-out pathway that does not delay or downgrade care.      | Spot-check audit of registration encounters confirming disclosure was given.                                                                                  |

|   |                                                                                                                                                                          |         |     |   |                                                                                                                                        |                                                                                                                 |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|---|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|   | informed this is happening.                                                                                                                                              |         |     |   |                                                                                                                                        |                                                                                                                 |
| 8 | <p>Deskilling — erosion of independent triage judgment over time.</p> <p>Long-term reliance on system suggestions could degrade nurses' own unaided triage accuracy.</p> | Ethical | L-M | H | Periodic “blind” triage audits where nurses triage without system input, to confirm skill retention against baseline.                  | Blind-audit ESI accuracy held stable over time relative to the pre-deployment baseline.                         |
| 9 | Training data demographic skew. Mercer's data is from one urban Caribbean ED; a model trained on it may not generalise to rural                                          | Equity  | H   | H | Deployment scope explicitly limited to validated sites; any new site requires local revalidation before go-live — mirroring Epic's own | Site-specific validation report is a hard gate before any new deployment — not a recommendation, a requirement. |

|    |                                                                                                                                                                                                                   |        |   |   |                                                                                          |                                                                                                           |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---|---|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|    | EDs or other island contexts.                                                                                                                                                                                     |        |   |   | corrective step (see Part 2).                                                            |                                                                                                           |
| 10 | Differential reliability across patient subgroups. Aggregate accuracy can mask poor performance for paediatric patients, elderly patients with atypical presentations, or non-standard dialect free-text capture. | Equity | M | H | Performance is reported by subgroup (age band, presentation type), never aggregate-only. | Subgroup-level sensitivity/specificity is published alongside overall figures in every validation report. |

## **Risk Memo**

### **1. Alert Fatigue (Risk 4)**

If the system cries wolf too often, nurses stop listening — including when it's right. This is not a hypothetical: it is the single most consistent failure mode found in the deployed clinical AI literature reviewed this week, most starkly in the Epic Sepsis Model, where clinicians had to investigate 109 alerts to find one genuine sepsis case. A Mercer system that floods triage nurses with low-value flags will be tuned out within days, regardless of how good the underlying logic is — and once nurses learn to ignore it, even a correct alert is at risk of being dismissed.

### **2. Distribution Shift / Single-Site Overfitting (Risk 1)**

A model is only as good as the population it was trained on. Mercer's eventual dataset will reflect one hospital, in one part of the Caribbean, over one stretch of time. If that model is later assumed to work identically at a different hospital — different demographics, different disease burden, different staffing — it can quietly fail in ways nobody notices until patients are harmed. This is exactly what happened with the Epic Sepsis Model: a tool validated in one context was deployed at hundreds of hospitals with wildly different patient populations, and its real-world performance fell far short of what internal testing had suggested.

### **3. Training Data Demographic Skew (Risk 9)**

This is the equity risk underneath the entire project's justification. The whole premise of the Phase 1 proposal is that AI triage tools built on North American and European data don't transfer to the Caribbean — but that same logic applies within the Caribbean too. A model trained only on Mercer's urban catchment will not automatically work for a rural clinic in a different territory with a different patient mix. Solving the “Caribbean data gap” with a single urban dataset risks just relocating the same generalisability problem to a smaller scale, rather than actually solving it. Obermeyer et al. (2019) sharpen this further: even within a single dataset, the choice of proxy variable can quietly encode the exact bias the project exists to avoid.

**Note:**

1. The Epic Sepsis Model case underlying Risks 1, 4, and 9 is committed separately as `docs/ai-harm-case-study.md` in the project repository.

## Proposed Solution

This project proposes a two-phase approach.

1. **Phase 1**, the 12-week pilot, focuses on designing and validating a structured digital triage capture tool aligned with the existing Mercer General paper triage form, producing a clean, de-identified, queryable dataset from real ED encounters.
2. **Phase 2**, beyond this programme, would use the developed dataset to train and locally validate a rule-based ESI classifier, iterating toward a machine learning model as data volume grows.

The Phase 1 deliverable is not an AI model. It is the dataset that makes a future model possible — and as Dr De Fretias has noted, this alone is genuinely valuable.

The tool would be designed for the nurse-led triage model at Mercer, with a minimal friction interface that respects the 3–5 minute triage window and does not increase documentation burden.

This week's risk analysis adds a forward-looking safeguard to that plan:

1. Phase 2 is not proposed as an open-ended commitment to build a classifier, but as a commitment to build one against a named, monitored set of risks.
2. Each risk in the register above carries a measurable signal precisely so that “it seems to be working” is never accepted as sufficient evidence on its own.

3. The Epic Sepsis Model case shows what happens when a tool is trusted on the strength of its developer's internal claims rather than independently and locally validated — this proposal commits, in writing, to not repeating that pattern.

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## Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

### Clinical & Vital Sign Terms

| Term | Meaning                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| ED   | Emergency Department                                                                                                               |
| ESI  | Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)                         |
| SBP  | Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)                                        |
| DBP  | Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)                                         |
| MAP  | Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as $(SBP + 2 \times DBP) / 3$ |
| GCS  | Glasgow Coma Scale — score from 3–15, measuring level of consciousness                                                             |
| RR   | Respiratory Rate — breaths per minute                                                                                              |

|      |                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------|
| FiO2 | Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)              |
| SpO2 | Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter                  |
| SBAR | Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication |

### Healthcare Systems & Infrastructure Terms

| Term          | Meaning                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| EHR           | Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal                  |
| LMIC          | Low- and Middle-Income Country — a World Bank income classification used in global health literature                                        |
| SIDS          | Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints                       |
| Boarding time | The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available |

## AI / Machine Learning & Risk Terms

| Term                  | Meaning                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| AI                    | Artificial Intelligence                                                                                                                |
| ML                    | Machine Learning                                                                                                                       |
| AUROC / AUC           | Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better                  |
| XGBoost               | Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data       |
| Rule-based classifier | A decision-making system using explicit if/then logic rather than a trained statistical model                                          |
| PPV                   | Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.        |
| Automation bias       | The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment     |
| Distribution shift    | When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on |
| Human-in-the-loop     | A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect   |

## Research & Documentation Terms

| Term              | Meaning                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------|
| DOI               | Digital Object Identifier — a permanent link format for academic papers                                            |
| APA               | American Psychological Association — the citation style used throughout this proposal (7th edition)                |
| Scoping review    | A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth |
| Systematic review | A more rigorous literature review following a defined search and inclusion protocol                                |